📘 Comprehensive Study Guide: Work, Energy, and Power

Level: Grade 12 Physics | **Unit:** Mechanics

1. Work (W): The Transfer of Energy

Work is a measure of energy transfer that occurs when a macroscopic force causes a displacement. Energy is used to do work, and the object receiving the work gains energy.

The General Formula (Constant Force)

For a constant force $\vec{F}$ acting over a rectilinear displacement $\vec{AB}$:

$$W_{\vec{F}} = \vec{F} \cdot \vec{AB} = F \cdot AB \cdot \cos(\alpha)$$

- F: Magnitude of the force in Newtons (N)
- AB: Magnitude of the displacement in meters (m)
- α: The angle between the force vector $\vec{F}$ and the displacement vector $\vec{AB}$.
- **Unit:** Joules (J). *(1 Joule = 1 N·m)*

Types of Work

Angle (α)	Sign of Work	Type of Work	Physical Meaning
$0^\circ \leq \alpha < 90^\circ$	$+$	**Motive Work**	The force *helps* the motion (adds energy).
$\alpha = 90^\circ$	0	**Null Work**	The force is perpendicular to motion (e.g., Normal force on a flat surface).
$90^\circ < \alpha \leq 180^\circ$	$-$	**Resistive Work**	The force *opposes* the motion (removes energy, e.g., friction).

Work Done by Weight ($W_{\vec{w}}$)

The work done by gravity is independent of the path taken and depends *only* on the change in vertical height (h):

$$W_{\vec{w}} = m \cdot g \cdot (h_{initial} - h_{final})$$

2. Energy (E): The Capacity to Do Work

The total energy of a system is the sum of its macroscopic **Mechanical Energy** (ME) and its microscopic **Internal Energy** (U).

$$E = ME + U$$

Note: In this chapter, we primarily focus on Mechanical Energy.

$$ME = KE + \Sigma PE = KE + GPE + EPE$$

A. Kinetic Energy (KE) — Energy of Motion

- **Translational KE**: For an object moving in a line or curve.

$$KE = \frac{1}{2}mv^2$$

- **Rotational KE** *(GS Track)*: For an object rotating about an axis.

$$KE_{rot} = \frac{1}{2}I\theta'^2$$

(Where I is the moment of inertia in $kg \cdot m^2$ and θ' is the angular speed in rad/s)

B. Potential Energy (PE) — Stored Energy

Energy stored due to an object's position, shape, or configuration.

1. **Gravitational Potential Energy (GPE)** Energy due to the interaction between the object's mass and the Earth.

$$GPE = mgh$$

⚠️ **Crucial Step:** You *must* define a Reference Level where $GPE = 0$ (usually the lowest point in the problem, like the ground).

- Above reference: $GPE > 0$
- Below reference: $GPE < 0$
- Relation to Work: $\Delta GPE = -W_{\vec{w}}$

2. **Elastic Potential Energy (EPE)** Energy stored in a compressed or stretched spring.

$$EPE = \frac{1}{2}kx^2$$

(Where k is the spring constant in N/m and $x = \Delta l$ is the deformation from the unstretched length in meters).

3. The Grand Theorems (Connecting Work & Energy)

Theorem 1: Work-Kinetic Energy Theorem

This is the most universal theorem. The total work done on a system by **ALL** external forces equals its change in kinetic energy.

$$W_{tot} = \Sigma W_{\vec{f}_{ext}} = \Delta KE = KE_f - KE_i$$

Theorem 2: Conservation of Mechanical Energy (Ideal Case)

Applies **ONLY** when an energy-isolated system has no non-conservative forces (like friction) doing work. Energy transforms perfectly between KE and PE.

$$\Delta ME = 0 \implies ME_i = ME_f$$

$$KE_i + GPE_i + EPE_i = KE_f + GPE_f + EPE_f$$

Theorem 3: Non-Conservation of Mechanical Energy (General Case)

When non-conservative forces (like friction $\vec{f}$ or a traction force $\vec{F}$) are present, the mechanical energy changes. The "lost" mechanical energy usually converts to Internal Energy (U), like heat.

$$\Delta ME = ME_f - ME_i = W_{\vec{f}} + W_{\vec{F}}$$

(If friction is the only non-conservative force, $\Delta ME = W_{\vec{f}}$, which is a negative value, meaning ME decreases).

4. Power (P): The Rate of Transfer

Power measures how fast work is done or how quickly energy is transferred.

Average Power (P_{avg})

The overall rate of work done over a given time interval Δt:

$$P_{avg} = \frac{W}{\Delta t} \quad \text{or} \quad P_{avg} = \frac{\Delta E}{\Delta t}$$

- **Unit**: Watts (W). *(1 Watt = 1 Joule/second)*

Relating Average Power to Force and Speed

If a constant force $\vec{F}$ acts on an object moving with an average velocity $\vec{v}_{avg}$ over a displacement $\Delta\vec{r}$, we can substitute the work formula ($W = \vec{F} \cdot \Delta\vec{r}$) directly into the power equation:

$$P_{avg} = \frac{\vec{F} \cdot \Delta\vec{r}}{\Delta t} = \vec{F} \cdot \left(\frac{\Delta\vec{r}}{\Delta t}\right) = \vec{F} \cdot \vec{v}_{avg}$$

If the force is applied in the exact same direction as the motion ($\alpha = 0°$), this simplifies to:

$$P_{avg} = F \cdot v_{avg}$$

Instantaneous Power (P)

While average power looks at a whole time interval, **instantaneous power** is the rate at which work is being done at one specific, exact moment in time.

By taking the time interval Δt to be infinitesimally small (approaching zero), we use the instantaneous velocity $\vec{v}$:

$$P = \vec{F} \cdot \vec{v}$$

If the force and instantaneous velocity are parallel:

$$P = F \cdot v$$

> 💡 **Application Note:** This formula is heavily used in vehicle and engine problems. For a car to maintain a constant speed v against resistive forces (like air resistance and friction f), the engine's traction force F must equal f. The engine must therefore continuously supply an instantaneous power of $P = F \cdot v$.

💡 The "5-Step" Problem Solving Framework

Whenever you face an energy problem, follow these exact steps:

1. **Identify the System**: What objects are you analyzing? (e.g., "System: block - Earth - spring").
2. **Define the Reference**: Explicitly write down where $GPE = 0$. Draw a dashed line on your diagram.
3. **Check for Non-Conservative Forces**: Is there friction, air resistance, or a motor force?
   - *No?* $\rightarrow$ Use **Conservation of ME** ($ME_i = ME_f$).
   - *Yes?* $\rightarrow$ Use **Non-Conservation of ME** ($\Delta ME = W_{nc}$) **OR** the **Work-KE Theorem** ($\Delta KE = W_{tot}$).
4. **List the States**: Define the initial (i) and final (f) positions. What is v, h, and x at each state?
5. **Solve**: Substitute your knowns into the chosen equation and isolate the unknown variable.